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Tomato: A Health Beneficial Vegetable Crop

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ABSTRACT

The tomato (*Solanum lycopersicum* L.) serves as an outstanding reservoir of minerals, nutrients, and secondary metabolites that significantly contribute to human well-being. The metabolites and antioxidant compounds present in tomato fruits play a crucial role in supporting human metabolism. Among these, lycopene, a naturally synthesized antioxidant abundantly found in tomatoes, stands out. Its presence in tomato fruits has proven beneficial in preventing various chronic diseases, including cardiovascular diseases and cancer. Tomatoes are also a vital source of other phytochemicals and bio active compounds with anti-mutagenic, anti-proliferative, anti-atherogenic, and anti-inflammatory properties. Essential vitamins like ascorbic acid and vitamin A, along with phenolic compounds such as flavonoids and phenolic acids, and carotenoids like α-carotene and β-carotene, as well as tomatine (glycoalkaloids), are present in substantial amounts. These phytochemicals and bio active compounds play a protective role by neutralizing free radicals responsible for the development of degenerative diseases in the body.

INTRODUCTION

Tomatoes, belonging to the *Solanum lycopersicum* L. species are a key component of the Mediterranean diet and widely consumed as vegetables due to their established health benefits. They are prominently featured in various processed food items like sauces, salads, soups, and pastes. Tomatoes are recognized for their nutritional richness, encompassing vitamins, minerals, fiber, protein, essential amino acids, monounsaturated fatty acids, carotenoids, and phytosterols. These nutrients play diverse roles in the body, contributing to constipation prevention, blood pressure reduction, blood circulation stimulation, lipid profile maintenance, detoxification, and the support of bone structure and strength. Beyond their nutritional content, tomatoes are a notable source of bio active compounds, or secondary metabolites, associated with preventing chronic degenerative diseases such as cardiovascular disease (CVD), cancer, and neurodegenerative conditions. The abundance of natural antioxidant chemicals in tomatoes, including carotenoids (β-carotenoids and lycopene), ascorbic acid (Vitamin C), tocopherol (Vitamin E), and bio active phenolic compounds (quercetin, kaempferol, naringenin, lutein, caffeic, ferulic, and chlorogenic acids), positions them as potential agents in mitigating various diseases, particularly chronic ones. These compounds exert their beneficial effects by inhibiting reactive oxygen species (ROS), scavenging free radicals, curtailing cellular proliferation and damage, preventing apoptosis, and influencing metal chelation, enzymatic activities, cytokine expression, and signal transduction pathways.

Lycopene, the primary carotenoid responsible for the red color of tomatoes, along with other phenolic compounds, exhibits pharmacological activities such as anticancer, anti-inflammatory, anti-diabetic, anti-allergenic, anti-atherogenic, anti-thrombotic, antimicrobial, antioxidant, vasodilator, and cardio-protective effects. In addition to their nutritive and health-promoting qualities, poly-phenolic compounds and carotenoids contribute to sensory aspects, including maintaining aroma, taste, and texture. Tomatoes are essential dietary sources of both soluble and insoluble dietary fibers, encompassing cellulose, hemicelluloses, and pectins. These fibers, resistant to intestinal digestion in the large intestine, are believed to alleviate various health issues, including bowel disorders, cancer, diabetes, CVDs, and obesity. Phytosterols, involved in preventing colon cancer and heart disease, are present in tomatoes, albeit in lower amounts compared to other fruits and vegetables. Notable phytosterols in tomatoes include β-sitosterol, campesterol, and stigmasterol. The antioxidant compounds in tomatoes primarily consist of various carotenoids, vitamin C, vitamin E, and phenolic compounds, conferring antioxidant activities by neutralizing reactive oxygen species (ROS) and protecting cell membranes against lipid peroxidation.

The nutritional composition of tomatoes can vary based on factors such as the cultivar, extraction procedures, analysis methods, and environmental conditions. During the processing of tomato products, up to 30% of their original weight may become waste, potentially retaining some nutritive values. For instance, tomato seeds and peel, the primary waste products, are rich in protein, dietary fibers, bio active compounds, and lycopene. Despite the numerous health benefits, excessive consumption of tomatoes may lead to undesired effects such as renal problems, allergies, arthritis, heartburn, and migraines.

Major bio-molecules in tomato

Tomato stands out as a remarkable member of the Solanaceous vegetable crops, renowned for its richness in phytochemicals, bio active compounds, plant pigments, minerals, and vitamins (Burton-Freeman, *et al.*, 2012). The important constituents of the tomato are presented here. The moisture content in a tomato fruit is approximately 95%, with the remaining 5% primarily composed of carbohydrates and fiber. Table-1 provides the nutritional value of 100 grams of tomato fruit.

Table 1: Raw materials and nutritive value of per 100 g of tomato fruit

Biomolecules	Nutritive contents	Value/100 g
Water	Water	94.5 g
Energy	Calories:	18 kcal 74 kJ
Carbohydrates	Total carbohydrates	3.9 g
	Sugars	2.6 g
Food fiber	Total food fiber	1.2 g
Vitamins	Vitamin–K	7.9 μg
	Vitamin–E	0.54 mg
	Vitamin–C	14 mg
	Vitamin–B6	0.08 mg
	Niacin (vitamin–B3)	0.594 mg
	Thiamine (vitamin–B1)	0.037 mg
	Retinol (vitamin–A)	42 μg
Proteins	Total proteins	0.9 g
Carotenoids	β-carotene	449 μg
Fat	Total fat	0.2 g
	Saturated:	0.03 g
	Monounsaturated:	0.03 g
	Polyunsaturated:	0.08 g
	Omega-	3: 0 g
	Omega-	6: 0.08 g
Minerals	Potassium:	237 mg
	Phosphorus:	24 mg
	Manganese:	0.114 mg
	Magnesium:	11 mg

Bioactive compounds

Bioactive compounds are naturally produced in a variety of fruits and vegetables, contributing to their health-promoting properties. These compounds often include antioxidants, anti-cancer agents, and anti-microbial substances (Ciccone *et al.*, 2013). In tomatoes, a key fruit in this context, the presence of significant bio active compounds brings about various health benefits for humans. The important bio active compound found in tomato in given in the table–2.

Table 2: Important bioactive compound of tomato and their health promotive properties

Major classes	Compounds	Health promotive properties
Phenolic compound	Phenolic acid	DNA oxidation protection and anti-tumor activity against carcinogenesis in the colon,
	Tannins	Antibacterial, antiviral, cardiovascular, anti- carcinogenic action, Adipogenesis inhibitors
	Flavonoids	Anti-inflammatory intestinal activity
Vitamins	Folates	Lowering the risk of neural tube defects, Monitoring homocysteine metabolism
	Vitamin E	Decreasing risks of advanced prostate cancer and type II diabetes, Inhibition of cardiovascular diseases and lipid peroxidation
	Vitamin C	DL oxidation inhibition, and monocyte adhesion
Carotenoids	β – carotene	Major source of vitamin – A, Photo-oxidant damageprevention
	Lutein	Eye health protection and symptom improvement in ARMD, improved endogenous damage and repair resistance to DNA
	Lycopene	Cancer inhibition (colorectal, prostate, breast, endometrial, oral, lung and pancreatic)

Lycopene and its health benefits

Tomatoes contain lycopene, a carotenoid that exists in the β-carotene form. The chemical composition of tomato-derived lycopene includes 11 conjugated and 2 non-conjugated components. (Cruz, *et al.*, 2018). Lycopene is polyunsaturated molecules feature 13 double bonds that can exist in either cis or trans configurations. Its significant antioxidant function enables it to react with oxygen molecules, effectively neutralizing free radicals. (Giovannucci, 1999). The transformation of green tomatoes to their characteristic red color is attributed to the action of lycopene. Numerous studies indicate that tomatoes possess a protective effect against various chronic diseases by mitigating oxidative stresses. Consumption of lycopene supplements in the diet has been associated with reduced rates of lipid oxidation, protein oxidation, as well as the oxidation of lipoproteins and DNA. Additionally, lycopene has been shown to have the potential to inhibit specific types of cancer cells within the human body. (Karppi, *et al*., 2013). Several reports have been demonstrated that higher lycopene concentration could be appropriate for the reducing risk of cardiovascular diseases and cancer.

Carotenoids

Carotenoids play a crucial role in both the process of photosynthesis and in providing protection against photo-damage (Lucock, 2000). With over 600 types found in nature, approximately 40 carotenoids are common in human diets. These compounds have the capacity to influence the expression of various proteins involved in cell proliferation and signaling pathways. Cyclin D_1, identified as an oncogene and commonly amplified in numerous breast cancer cell lines, has been extensively studied (Baranska *et al*., 2006). Lycopene, a component found in tomatoes, is linked to a reduction in Cyclin D_1 protein levels. Additionally, lycopene has been observed to enhance the expression of various proteins associated with cellular differentiation, including cell surface antigen (CD14), reactive oxygen species, and receptors for chemo-tactic peptides (Luthria *et al*., in 2006). The crucial nutritional contribution of tomatoes lies in their carotenoid content, particularly lycopene, making carotenoid intake the most significant factor in the Antioxidant Nutritional Quality Index.

Tomato fruits contain essential carotenoids, including β-carotene, which serves as a precursor for vitamin A biosynthesis. Notably, β-carotene in tomatoes offers distinct advantages compared to other carotenoids present in the fruit (Beauvoit *et al.*, in 2014). Research in epidermal studies indicates that a sufficient intake of β-carotene can reduce the risk of cancer and prevent various chronic disorders. Additionally, dietary consumption of tomato paste has been shown to provide protection against erythema in humans, with a demonstrated safeguarding effect against UV radiation when applied to the skin over a 10-week period. Numerous studies have reported that the carotenoids present in tomatoes possess antioxidant properties capable of stimulating the immune system (Maiani *et al.*, 2009). The absorption of carotenoids primarily occurs in the small intestine, serving as a crucial source of vitamin E in the form of alpha-tocopherol. Moreover, the nutritional value of tomatoes is further enriched by the presence of poly-phenolic compounds, flavonoids, and carotenoids, enhancing their overall health benefits (Pem and Jeewon in 2015).

Phenolic Compounds and flavonoids

Phenolic acids and flavonoids are secondary plant products produced by plants, serving a role distinct from the primary functions related to growth and development (Kumar and Pandey 2013). The primary purpose of synthesizing these compounds in plants is for protection against various biological threats. Fortunately, these naturally occurring compounds have medicinal and pharmaceutical properties that benefit human health. Tomatoes, for instance, naturally synthesize various phenolic compounds and flavonoids, such as phenolic acid, flavonoids, hydroxycinnamic acid, coumarins, lignans, tannins, and gallic acid. In tomatoes, phenolic compounds, along with their lycopene content, contribute to antioxidant activity (Ghasemzadeh and Ghasemzadeh, 2011). These compounds exist in both insoluble-bound and free soluble forms. The insoluble-bound form is associated with protective functions, such as shielding the skin from UV radiation and pathogens (Peralta *et al*., 2007). The quantity and composition of phenolic compounds are influenced by various factors, including unfavorable environmental conditions, as well as the presence of insects, pests, and diseases. These factors can impact the overall quality and quantity of phenolic compounds in tomatoes.

Vitamins

Tomato fruits are a source of essential vitamins, including vitamin C, vitamin E, and vitamin A, as well as containing various other compounds such as sucrose, hexose, citrate, malate, and ascorbic acid (Porrini *et al*., 2005). The vitamin C present in tomatoes serves as a powerful antioxidant, offering protection to body cells against cancer. Tocopherol, another component in tomato fruit, plays a role in sustaining photosynthesis, particularly in challenging conditions like high temperatures and water deficit. It has been observed in the various studies, the uncooked tomato fruits are more nutritious than cocked (Prasad, *et al*., 2011). Some studies revealed that the potential benefits of tomato fruits in treating hypertension and managing blood pressure, attributing these effects to their systolic properties and antioxidant activity. Notably, the presence of ascorbic acid in tomatoes has been identified as a key factor in these health benefits. (Raiola *et al*., 2014) observed that the levels of ascorbic acid in tomato fruits increase gradually and then decline upon the initiation of fruit ripening, reaching a certain threshold. The reduction in ascorbic acid during tomato fruit ripening is associated with a noticeable color change, coinciding with an elevation in the activity of the enzyme ascorbic oxidase (Ranjan *et al*., in 2012). Investigations related to ascorbic acid indicated that free radicals in the body cells neutralized due to sufficient use of tomato derived ascorbic acid.

CONCLUSION

Tomatoes are widely consumed either in their fresh form or as processed products, making them the second most consumed vegetable globally and a crucial source of nourishment for the world's population. The vegetable is known for its richness in bio active compounds, including lycopene, carotenoids, β-carotene, lutein, and vitamins, all of which contribute to various health-promoting effects. Recently, tomatoes have gained recognition due to the presence of numerous phytochemicals and bio active compounds that hold known health benefits for the human body. The carotenoids and lycopene found in tomato fruits have been identified as active substances that may play a role in preventing the development of cancer cells in the human body. However, it has also been noted that consuming tomatoes can lead to various types of allergic reactions due to the presence of allergenic proteins in the fruits. Addressing the challenge of enhancing tomato fruit quality and reducing the levels of undesirable proteins is a complex issue. In the future, the strategic application of biotechnological methods and gene editing techniques may prove beneficial in improving tomato crops.

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